

Flash Forays

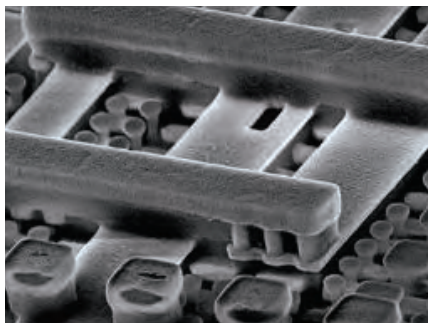
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In 1984, Dr. Fujio Masuoka of Toshiba announced flash to the world. Today, flash memory is everywhere – pen drives, thumb drives, MP3/MP4 players, digital cameras, PDAs, and virtually all portable electronic devices that need to store data. In fact, they're slated to replace hard disk drives pretty soon.

Inside a flash drive

Flash drives basically store data on an electronically erasable programmable read only memory (EEPROM) chip. This has a lattice of columns (bitlines) and rows (wordlines). Each point of intersection is a cell. A typical flash chip contains billions of such cells. Each cell is made of a transistor

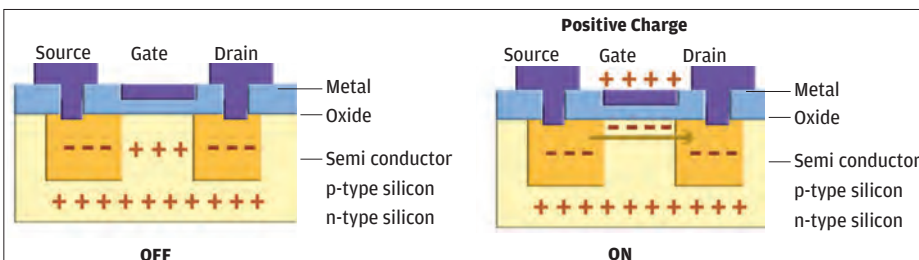


Peeking into a Flash Drive

– a MOSFET, to be precise – which has an additional gate called a 'floating' gate added on to it.

MOSFETs

A MOSFET (Metal Oxide Semiconductor Field Effect Transistor) is midway between a diode (two p and n type semiconductor layers) and a transistor (sandwiched layers of p-type and n-type semiconductor layers). An n-type semiconductor releases electrons more readily, while p-type semiconductors allow electrons to flow into them easily. The MOSFET, however, is set up a little differently from the typical bipolar diode with a base coated with a layer of metal and oxide. The holes in the p-type layer prevent the excess electrons from passing from one n-type semiconductor (source) to the other (drain). But if we just let a positive voltage flow in from the gate, this creates an electric field. Now electrons start moving through



How MOSFETs work

the channel from the source to the drain. This makes it *unipolar* because a single electric charge is enough to get the electrons moving.

The difference between the flash memory transistor and the typical MOSFET is the floating gate. Flash drives use FG MOS cells (Floating Gate Metal Oxide Semiconductor FETs) that are basically MOSFETs with an additional gate. This is an electrically isolated gate, with one or more secondary gates deposited above it, also electrically isolated from it. This floating gate is surrounded by resistive material. This way it is difficult for electrons to pass into the gate. Once they have entered, it also ensures they stay trapped until the bias is reversed. If

you apply a positive potential to the control gate, electrons (being negatively charged) are thrust through the layer of oxide (usually silicon dioxide) between the p-type substrate. Some of these are pushed into the floating gate and the electrons are then trapped there until forced out again.

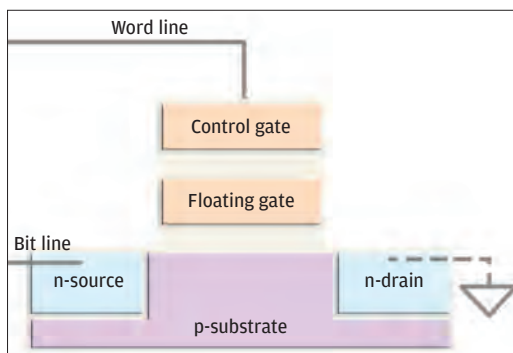
The electrons enter the floating gate in two ways:

- **Fowler-Nordheim Tunneling:** This is a phenomenon where electrons are emitted from a semiconductor into a dielectric or a vacuum when persuaded by a powerful electric field. Unlike other forms of emission, electrons pass through a barrier – instead of working around it – because it has not yet lost its wave function and, decaying into the barrier, thus heightening the probability of it emerging on the other side.

- **Hot Carrier Injection:** If an electron or a 'hole' (an absent electron) in a semiconductor can be made to gain sufficient kinetic energy to pass through a barrier, the process is known as 'Hot Carrier Injection'.

NAND and NOR flash Memory

The two most common kinds of flash memory are NAND and NOR. The terms are from Boolean logic and



Parts of an FG MOS transistor

HOW TO PROTECT YOUR FLASH DRIVE

Never lost your pen drive or got a virus infection in it? You're kidding me. Given that they're featherweight and getting tinier by the day, chances of your flash memory device getting into the wrong hands are increasing day by day. And everybody knows how the 'Autoplay' function in Windows gets trojans and worms into the works. Here are a few tips to safeguard your flash data:

1. Buy a write protected flash drive: Once a flash drive is write protected, data will only get written if you disable the 'write protect' function.

2. Put in a password: There's plenty of freeware on the web which allows you to password protect your flash drive, thus ensuring that only you have access to the data in it.

3. Encrypt the data: If you do this, even after somebody cracks your password, the data will only look like gobbledygook to him.

4. Anti-virus: Clean your flash drive of malicious software every time you plug it into a computer, using reliable anti-virus software.